

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-019

WASTE DISCHARGE REQUIREMENTS
FOR
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Hidden Valley Lake Community Services District (District) submitted a Report of Waste Discharge, dated 28 May 1999, requesting approval to reuse stabilized sludge (biosolids) from its wastewater treatment plant. Supplemental information, including a Draft Supplemental Environmental Impact Report, was received on 23 August 1999.
2. The Hidden Valley Lake Association uses treated effluent to spray irrigate the community golf course within the Hidden Valley Lake Subdivision. The golf course is approximately 135 acres, and is owned, operated, and maintained by the Hidden Valley Lake Association.
3. Hidden Valley Lake Community Services District and the Hidden Valley Lake Association are hereafter jointly referred to as "Discharger".
4. The Board, on 20 May 1994, adopted Order No. 94-138 which prescribed requirements for the District's activated sludge-extended aeration treatment plant and recycled water spray irrigation practices.
5. Order No. 94-138 does not specifically prescribe requirements for the application of biosolids to a supplemental spray irrigation field owned by the District, and is not consistent with current plans and policies of the Board.
6. The existing wastewater collection system, activated sludge-extended aeration treatment facility, recycled water spray irrigation and biosolid application areas (Assessor's Parcel Nos. 14-270-10 and 14-280-16) are owned by the Discharger in Lake County. As shown in Attachment A, these areas are found within the projected Section 30, T11N, R6W, MDB&M. The golf course disposal area is in projected Sections 17, 18, 19, 20 & 21, T11N, R6W, MDB&M. Local surface waters, the treatment facility, and the recycled water, biosolids application and wastewater disposal areas and are shown in Attachment B. Attachments A and B are attached hereto and part of the Order by reference.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-2-

7. The existing treatment facility became operational on 15 February 1996 and includes an activated sludge-extended aeration plant with an average dry weather flow of 0.350 million gallons per day (mgd), and a peak wet weather flow of 0.894 mgd. The facility processes include primary screening, secondary treatment through an activated sludge and clarification process, chemical addition, direct tertiary filtration, chlorination, six sludge-drying beds, a concrete-lined equalization basin and an effluent storage basin. Treated effluent is stored in the 412-acre-feet, clay-lined, effluent storage basin during periods when irrigation is prohibited. Monitoring wells are in place at the treatment site to measure any potential effect of the impounded wastewater to groundwater.
8. The Discharger uses treated effluent to spray irrigate the community golf course located within the Hidden Valley Lake Subdivision, as well as a supplemental spray irrigation field (supplemental field) located between the treatment plant site and Grange Road. The supplemental field contains approximately 45 acres of suitable area for reuse. The Discharger will also use the 45-acre supplemental field for the application of up to 200 tons of biosolids per year.
9. A ground water study was conducted by the Discharger to model the occurrence of nitrates in the ground water as the result of seepage from the bottom of the storage pond. The study concluded that an effluent storage pond liner of 18" thickness and 1×10^{-6} cm/sec permeability, or equivalent thickness and permeability, would have minimal impact on ground water quality. The ground water study was based upon conservative assumptions regarding various aspects affecting the potential for ground water contamination. These assumptions included: 1) using 100-year storm event data modeled for 30 consecutive years, 2) neglecting biological denitrification processes at the pond bottom, 3) assuming that background nitrate levels in ground water are negligible, and 4) using 1×10^{-6} cm/sec as overall permeability, whereas actual constructed liner permeabilities should be less than 1×10^{-6} cm/sec in some areas.
10. Recycled water is delivered from the effluent storage basin to two irrigation ponds located on the golf course. Irrigation flows are proposed to reach a maximum of 1.3 mgd, weather permitting.
11. According to the 28 May 1999 Report of Waste Discharge, at an ultimate design capacity of 0.350 mgd, up to 200 tons of biosolids will be produced from the treatment plant per year. Waste solids generated from the biological and filtration processes are dewatered in sludge drying beds and will be applied to the designated supplemental field. Due to required setbacks and buffer zones, the available application area will be limited to approximately 45-acres. The setback distances are incorporated in this Order.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-3-

12. As described in the Discharger's 26 January 1993 Report of Waste Discharge, characteristics of the treated wastewater are:

<u>Constituents</u>	<u>Concentrations</u>
BOD ₅	30 mg/l
Total Suspended Solids	30 mg/l
Total Coliform	2.2 MPN/100ml Median
Turbidity	2 NTU

13. Local annual precipitation averages 40 to 45 inches. The mean evaporation rate is approximately 45 inches per year. The treatment facility and the adjacent supplemental field are partially within the 100-year floodplain.
14. The United States Environmental Protection Agency (USEPA) has promulgated biosolids reuse regulations in 40 CFR 503, *Standard for the Use or Disposal of Sewage Sludge*, which establishes management criteria for protection of ground and surface waters, sets application rates for heavy metals, and establishes stabilization and disinfection criteria.
15. The Board is using the Standards in 40 CFR 503 as guidelines in establishing this Order, but the Board is not the implementing agency for 40 CFR 503 regulations. The Discharger may have separate and/or additional compliance, reporting, and permitting responsibilities to USEPA, which are not covered by this Order.
16. The following are the Discharger's metals concentrations in the biosolids as reported in the 28 May 1999 Report of Waste Discharge:

<u>Constituents</u>	<u>Concentrations</u>	<u>Units¹</u>
Arsenic	<0.5	mg/kg
Cadmium	<10	mg/kg
Chromium	90	mg/kg
Copper	680	mg/kg
Lead	<25	mg/kg
Mercury	<0.1	mg/kg
Molybdenum	<25	mg/kg
Nickel	29	mg/kg
Selenium	<0.1	mg/kg
Zinc	306	mg/kg

¹Results are reported in dry weight

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-4-

17. The Discharger reports that acceptable biosolids will have been treated to meet the standards for a process to significantly reduce pathogens as defined by 40 CFR 503.
18. The biosolids application rates will vary but will be established based on crop utilization and limitations imposed by annual and/or cumulative metal loading criteria. Biosolids/soil mixture will not be allowed to exceed maximum soluble metal concentrations to protect groundwater. Metals concentration in the recycled water will also have to be evaluated and taken into account in calculating the total metals loading and soluble metal loading to the soil. Crops will be limited to pasture grasses for livestock grazing and fodder crops.
19. The application of biosolids will occur annually on approximately 45 acres of the designated supplemental field. Biosolids will be applied using spreading equipment and will be incorporated into the soil within 24 hours of initial biosolids application. Any new application site in conjunction with the Discharger's submittal of an updated Biosolids Management Plan will have to be approved by the Executive Officer prior to application.
20. Based on a Soil Conservation Service Soil Survey, the soil in the proposed application area is underlain by the Lupoyoma silt-loam on the north end, and the Maxwell and Yorkville Variant clay-loams on the south side. Permeability rate for the clay and clay loam soils encountered at the site were below 6.0 inches/hour.
21. The treatment plant and supplemental field site are sensitive due to the proximity to both individual and the Discharger's domestic water supply wells, and to within 1000 feet of residential homes. Ground water has been observed as shallow as 4.5 feet below existing grade throughout the Dischargers property. The Discharger's three drinking water wells are located just north of Grange Road, near the wastewater treatment and disposal facility and the supplemental field. These wells draw water from the deeper zones of 80 and 170 feet. A drainage ditch extends in an easterly direction across the supplemental field. Drainage from this area flows south to Crazy Creek, then north and east in Crazy Creek to Putah Creek.
22. Surface water drainage is to Crazy Creek, then to Putah Creek and ultimately to Lake Berryessa.
23. The State Water Resources Control Board adopted Order No. 97-03-DWQ (General Permit No. CAS000001) specifying waste discharge requirements for discharges of storm water associated with industrial activities, excluding construction activities, and requiring

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-5-

submittal of a Notice of Intent by industries to be covered under the permit. However, this Order incorporates specific limitations and prohibitions to mitigate potential water quality problems associated with storm water runoff.

24. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
25. The beneficial uses of Lake Berryessa and Putah Creek are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; ground water recharge; fresh water replenishment; hydroelectric power generation; and preservation and enhancement of fish, wildlife and other aquatic resources.
26. The beneficial uses of the underlying groundwater are municipal, industrial, and agricultural supply.
27. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations, Section 60301, et seq. (hereafter Title 22) for the use of recycled water and has developed guidelines for biosolids reuse in the *Manual of Good Practices for Landspreading of Sewage Sludge*, April 1993. The Discharger's operation and management of its biosolids program will be consistent with these guidelines as well as the USEPA's biosolids reuse regulations in 40 CFR 503, *Standard for the Use or Disposal of Sewage Sludge*.
28. Prior to the issuance of Order No. 94-138, the Discharger adopted a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The EIR indicated that the wastewater treatment facility, recycled water irrigation and biosolids application practices may have the following significant impacts on water quality:
 - a. Increase in turbidity of surface water and streams during and after construction.
 - b. Water quality effects from force main rupture or pump station failure.
 - c. Potential for contamination of surface water by sewage during flooding or damage to treatment plant and disposal facilities from flooding.
 - d. Pollution of groundwater by infiltration of storage pond effluent.
 - e. Potential for contamination of ground water with excess nutrients, particularly

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-6-

nitrogen, from irrigation with recycled water at the supplemental irrigation field located between the treatment plant site and Grange Road.

29. In 1994, the Board has reviewed the EIR and determined that compliance with the waste discharge requirements will mitigate or avoid the significant impacts on air and water quality listed in Finding No. 28. Specific mitigation measures correspond to the alphabetically listed impacts in Finding No. 28 and are as follows:
- a. i. Control increased turbidity of surface water by using standard erosion control measures.
 - b. i. The pump station and transmission main system should be designed for leak detection and early warning to hasten necessary repairs.
ii. The transmission main crossing of Putah Creek should be encased in concrete to minimize the potential for rupture.
iii. The terminal pump station and treatment plant shall be provided with emergency power supplies.
 - c. i. Irrigation systems located in the floodplain shall be valved to isolate those areas in the floodplain from irrigation.
ii. All pumps and facilities shall be flood-proofed to one foot above the 100-year floodplain.
iii. The facility will be equipped with a portable generator as a source of power in the event of an emergency.
 - d. i. The effluent storage pond will be fitted with the proper liner as geotechnical/engineering studies of the site indicate that a liner is necessary to mitigate nitrate leakage into the ground water.
 - e. i. The treatment process will be modified to reduce nitrate concentrations that may adversely affect the water quality in wells located along Grange Road when the supplemental field is in use.
ii. When use of the supplemental field commences, ongoing monitoring studies will be conducted to demonstrate that no domestic water supply

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-7-

quality is being degraded. Should a potential for degradation of potable water supplies be demonstrated, the treatment/storage/disposal process will be modified to eliminate the potential for degradation.

30. This discharge is exempt from the requirements of Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste, as set forth in Title 27, California Code of Regulations (CCR), Division 2, Subdivision 1, Section 20005, et seq. (hereafter Title 27). The exemption, pursuant to Section 20090 (b), is based on the following:
 - a. The Board is issuing waste discharge requirements, and
 - b. The discharge complies with the Basin Plan, and
 - c. The biosolids to be applied are a non-hazardous, decomposable waste applied as a soil amendment pursuant to best management practices, and
 - d. The wastewater does not need to be managed according to CCR, Title 22, Division 4.5, Chapter 11, as a hazardous waste.
31. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
32. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 94-138 is rescinded and Hidden Valley Lake Community Services District and Hidden Valley Lake Association, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-8-

3. The discharge of waste classified as 'hazardous' under CCR, Title 23, Chapter 15, Section 2521, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
4. The application of biosolids at rates in excess of the nitrogen requirements of the vegetation, or at rates that would cause the excess nitrogen or metals to leach to groundwater, is prohibited.
5. The application of biosolids shall be confined to the designated application areas, as stated in Finding Nos. 6, 8 and 11 of this Order and shown on Attachment B.
6. The grazing of milking animals used for producing unpasteurized milk for human consumption is prohibited on the designated recycled water irrigation or biosolids application areas.
7. The off-site discharge of storm water, and/or irrigation runoff, from fields on which biosolids have been applied, is prohibited for 30 days following the application of biosolids.
8. The application of biosolids during rainfall events is prohibited.
9. The application of biosolids with pollutant concentrations greater than those shown below is prohibited.

<u>Constituents</u>	<u>Ceiling Concentration</u> <u>mg/kg dry weight</u>
Arsenic	75
Cadmium	85
Chromium	3000
Copper	4300
Lead	840
Mercury	57
Molybdenum	75
Nickel	420
Selenium	100
Zinc	7500

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-9-

B. Discharge Specifications:

Treatment Plant and Wastewater Storage Facilities

1. The average monthly dry weather discharge from the treatment plant shall not exceed 0.350 mgd.
2. The maximum wet weather discharge from the treatment plant shall not exceed 0.894 mgd.
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment, recycled water irrigation, biosolids application or wastewater disposal areas.
4. As a means of discerning compliance with Discharge Specification No. 3, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
5. The treatment and storage facilities shall be designed, constructed, operated and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
6. The effluent from the treatment facility shall not exceed the following limits:

Constituent	Units	Monthly Average	Daily Maximum
BOD ₅ ¹	mg/l	30	60
Suspended Solids	mg/l	30	60
Settleable Solids	ml/l	-	0.5
Coliform	MPN/100ml	2.2 Median	23
Turbidity	NTU	2	5

¹Five-day, 20°C biochemical oxygen demand (BOD).

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-10-

7. Treatment plant effluent shall not have a pH less than 6.5 or greater than 8.5.
8. The storage pond shall have a clay liner a minimum of 18" thick, installed at a relative compaction of at least 90%, with a maximum permeability of 1×10^{-6} cm/sec, or equivalent.
9. Ponds shall be managed to prevent breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
10. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
11. Ponds shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).
12. On or about **1 October each year**, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification No. 11.

Recycled Water Irrigation

13. A maximum of 1.3 mgd may be discharged to the golf course and supplemental field, weather permitting.
14. There shall be no standing water in the disposal area 48 hours after wastewater is applied.
15. Areas irrigated with recycled water shall be managed to prevent breeding of mosquitoes. More specifically,

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-11-

- a. Tail water must be returned, and applied irrigation water must infiltrate completely within a 48-hour period.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store recycled water.
16. Recycled water for irrigation shall be managed to minimize erosion, runoff, and movement of aerosols from the disposal areas.
17. Direct or windblown spray shall be confined to the designated recycled water spray irrigation areas (appropriate portions of the golf course and the supplemental field) and prevented from contacting drinking water facilities.
18. The Discharger may not spray irrigate recycled water during periods of precipitation and for at least 24 hours after cessation of precipitation, or when winds exceed 30 mph.
19. The discharge shall remain within designated disposal areas at all times.
20. Signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal to alert the public of the use of recycled water.
21. Golf course scorecards should indicate that recycled water is being used for irrigation purposes.
22. Storm water runoff from the irrigation areas shall not be discharged to any surface water drainage course within 48 hours of the last application of recycled water.
23. Recycled water irrigation practices shall be in accordance with Winzler and Kelly Consulting Services' September 1993 *Engineering Report on the Production Distribution and Use of Reclaimed Water*. Application of recycled water to the reclamation areas shall be at reasonable rates considering the crop, soil, climate, and irrigation management system. The nutrient loading of the reclamation areas, including the nutritive value of organic and chemical fertilizers and of the recycled water shall not exceed the crop demand.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-12-

24. There shall be no impoundment of recycled water, recycled water irrigation or biosolids application within 500 feet of any domestic water well, or within 100 feet of any irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that less distance is justified.

Biosolids Application

25. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with CCR, Title 27, and approved by the Executive Officer.
26. Prior to biosolids application, the Discharger shall submit a Biosolids Management Plan for approval by the Executive Officer. The Discharger shall maintain and operate under this approved Biosolids Management Plan.
27. Biosolids shall comply with either Class A or Class B pathogen standards as listed in Appendix B of 40 CFR 503.
28. Biosolids shall comply with one of the vector attraction reduction standards as listed in 40 CFR 505.33.
29. Biosolids shall be landspread and incorporated into the soil within 24 hours of arrival at the disposal fields. If the Vector Attraction Reduction alternative 40 CFR 503.33 (b)(10)(i) is used, the biosolids must be incorporated into the ground within six (6) hours of landspreading.
30. Biosolids application rates shall not exceed crop utilization rates, or rates which cause specific constituents to exceed single, annual, or lifetime application limits based on:
 - a. 40 CFR 503, Standard for the Use or Disposal of Sewage Sludge;
 - b. Soil Cation Exchange Rates;
 - c. Soil pH;
 - d. Nitrogen demand of the crop; and
 - e. Phytotoxicity.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-13-

31. Biosolids shall not be applied to soil with a pH of less than 6.5
32. Biosolids shall not be applied to water-saturated, snow-covered or frozen ground.
33. Biosolids shall not be applied to slopes exceeding 10 degrees, unless a certified agronomist provides sufficient information in the Pre-Application report outlining site conditions that justify application on steeper slopes. Furthermore, application to slopes exceeding 10 degrees shall not be allowed from **30 October through 30 April**. Biosolids applied to slopes 10 degrees or greater shall be incorporated immediately. Contour tilling shall be required on slopes greater than 10 degrees. In no case shall biosolids application be made on slopes exceeding 28 degrees.
34. Objectionable odors originating at this facility, as a result of applying biosolids, shall not be perceivable beyond the limits of the property owned by the Discharger.
35. Biosolids shall not be applied to land in amounts which cause the following cumulative loading rates to be exceeded:

<u>Constituents</u>	<u>Cumulative Loading Rate (kg per hectare)</u>
Arsenic	41
Cadmium	39
Chromium	3000
Copper	1500
Lead	300
Mercury	17
Molybdenum	18
Nickel	420
Selenium	100
Zinc	2800

36. To protect groundwater, the application of biosolids shall not cause the biosolids/soil mixture to exceed the following soluble levels (mg/l) when extracted with distilled water using the Waste Extraction Test (WET) described in the CCR, Title 22, Division 4, Chapter 30:

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-14-

Concentrations in Biosolids/Soil Mixture Not to be Exceeded

Metal	Limitation (mg/l)
Cadmium	0.1
Copper	20.0
Lead	0.5
Nickel	0.134
Zinc	200.0

37. After the last application of biosolids in the supplemental field, the Discharger shall ensure the following:
- a. For at least 30 days:
 - 1) Public access to the application site is restricted for land with a low potential for public exposure;
 - 2) Food, feed, and fiber crops are not harvested; and
 - 3) Grazing by animals is prevented.
 - b. For at least 14 months:
 - 1) Public access to the site is restricted for land with a high potential for public exposure;
 - 2) Turf is not to be harvested if the harvested turf is to be placed on land with a high degree of public exposure; and
 - 3) Grazing of milking animals used in the production of unpasteurized milk for human consumption is prevented.
 - c. For at least 14 months:

Food crops with harvested parts that touch the biosolids/soil mixture and are completely above the land surface are not harvested.
 - d. For at least 36 months:

Prevent planting of unprocessed food chain crops.
 - e. For at least 38 months:

Food crops with harvested parts below the land surface are not harvested, unless the biosolids remained exposed on the ground surface for at least four months prior to incorporation into the soil.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-15-

38. Staging areas, storage areas, and biosolids application shall be at least:
 - a. 25 feet from property lines;
 - b. 500 feet from water supply wells or occupied dwellings;
 - c. 50 feet from public roads;
 - d. 100 feet from surface waters, including rivers, creeks, ponds, lakes, marshes, and their tributaries;
 - e. 10 feet from agricultural dwellings; and
 - f. 100 feet from residential developments.
39. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.

C. Biosolids Storage Specifications:

Biosolids shall be considered "stored" if they are placed on the ground or in non-mobile containers (i.e., not in truck or trailer) at the application site or an intermediate storage location away from the generator/processing site prior to application. "Storage" does not include biosolids placed on the ground for brief periods of time solely to facilitate transfer the biosolids between transportation and application vehicles.

1. Biosolids shall not be stored for more than seven consecutive days prior to application.
2. Biosolids containing free liquids shall not be placed on the ground prior to application on an approved site.
3. Sites for the storage of biosolids shall be located, designed, and maintained to restrict public access to the biosolids.
4. Biosolids shall not be stored directly on the ground at any one location for more than seven days in any 60-day period.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-16-

5. Biosolids storage sites which contain biosolids between 1 October and 30 April shall be designed and maintained to prevent washout or inundation from a storm or flood with a return frequency of 100-year.
6. Biosolids storage facilities which contain biosolids between 1 October and 30 April shall be designed and maintained to contain all storm water falling from a 10-year, 24-hour storm.
7. Biosolids storage facilities shall be designed, maintained, and operated to minimize the generation of leachate.
8. If biosolids are to be stored at the site, a plan describing the storage program and means of complying with this Order shall be submitted for Executive Officer approval no later than 60-days prior to the storage of biosolids. The storage of biosolids shall not commence until after approval of the plan.
9. The Discharger shall operate the biosolids storage facilities in accordance with the approved Biosolids Management Plan required in Discharge Specification B.26.

D. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number of total coliform organisms to exceed 2.2/100 ml over any 7-day period.

E. Provisions:

1. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that are essentially free of pollutants.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-17-

2. The Discharger shall use the best practical cost-effective control technique currently available to limit mineralization to no more than a reasonable increment.
3. The Discharger shall implement the necessary legal authorities, programs, and controls to ensure that the following incompatible wastes are not introduced to the treatment system, where incompatible wastes are:
 - a. Wastes which create a fire or explosion hazard in the treatment works;
 - b. Wastes which will cause corrosive structural damage to treatment works, but in no case wastes with a pH lower than 5.0, unless the works is specially designed to accommodate such wastes;
 - c. Solid or viscous wastes in amounts which cause obstruction to flow in sewers, or which cause other interference with proper operation or treatment works;
 - d. Any waste, including oxygen demanding pollutants (BOD, etc.) released in such volume or strength as to cause inhibition or disruption in the treatment works, and subsequent treatment process upset and loss of treatment efficiency;
 - e. Heated in amounts that inhibit or disrupt biological activity in the treatment works, or that raise influent temperatures above 40°C (104°F), unless the treatment works is designed to accommodate such heat;
 - f. Petroleum oil, nonbiodegradable cutting oil, or products of mineral oil origin in amounts that will cause interference or pass through;
 - g. Pollutants which result in the presence of toxic gases, vapors, or fumes within the treatment works in a quantity that may cause acute worker health and safety problems; and
 - h. Trucked or hauled pollutants, unless predesignated by the Discharger.
4. For the purposes of this Order, "spray irrigation" means application of recycled water to crops by sprinklers.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-18-

5. Recycled water controllers, valves, quick couplers, sprinkler heads etc., shall be affixed with recycled water warning signs and shall be of a type, or secured in a manner, that permits operation by authorized personnel only.
6. Piping, valves and outlets shall be marked to differentiate recycled water from domestic water supply.
7. Supplementing recycled water by connection with a domestic drinking water source or irrigation or industrial wells requires an air gap device.
8. A contingency plan, including notification of the Board and health agencies, and outlining action to be taken when effluent quality fails to meet required standards, shall be submitted within 60 days of adoption of this Order.
9. If the Discharger intends to use recycled water on crops other than those specified in an accepted land management plan, it shall first submit a written report demonstrating, to the satisfaction of the Executive Officer, that management of recycled water and irrigated properties will assure compliance with the terms of this Order.
10. Neither the distribution or application of biosolids shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
11. The Discharger shall comply with the Monitoring and Reporting Program No. 5-00-019, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
12. The Discharger shall submit a report with an updated Biosolids Management Plan prior to the biosolids application to the designated 45-acre supplemental field. The report must demonstrate Discharger's ability to comply with this Order. In addition, this report will include, but not be limited to, a monitoring program for the ditches and stream for the proposed application site. This report may also be included with the Pre-Application Report required in Monitoring and Reporting Program No. 5-00-019.
13. The Discharger shall not apply biosolids until all permitting requirements for Lake County have been satisfied.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

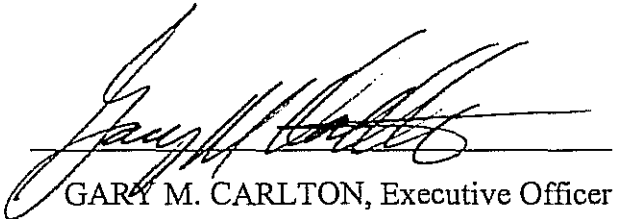
-19-

14. Biosolids reuse authorized by this Order is limited to stabilized wastewater treatment plant biosolids approved of by the Executive Officer. The biosolids must be treated as described in Finding No. 17.
15. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
16. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
17. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or application areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
18. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
19. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
20. If recycled water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of recycled water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with CCR, Title 22, Division 4.
21. The Board will review this Order periodically and will revise requirements when necessary.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-00-019
HIDDEN VALLEY LAKE COMMUNITY SERVICES DISTRICT
HIDDEN VALLEY LAKE ASSOCIATION
HIDDEN VALLEY LAKE WATER RECLAMATION FACILITY
LAKE COUNTY

-20-

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 28 January 2000.



GARY M. CARLTON, Executive Officer

AMENDED

c:\my documents\job files\lake\hidden valley\990923hv.wdr3.doc

SKC:1/28/00